Q13. Arduino Pin Types

1. Digital Input

Definition:
Digital input pins are used to read digital signals. They can detect only two states: HIGH (1) and LOW (0).

Example: Push Button

IoT Use Case:
A smart door lock uses a push button as a digital input to detect when the user presses the button to unlock the door.

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2. Digital Output

Definition:
Digital output pins are used to send HIGH or LOW signals to control external devices.

Example: LED

IoT Use Case:
A smart home automation system uses digital output pins to switch lights or relays ON and OFF.

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3. Analog Input

Definition:
Analog input pins read continuously changing voltage values and convert them into digital values ranging from 0 to 1023.

Example: LDR (Light Dependent Resistor)

IoT Use Case:
A smart street lighting system uses an LDR sensor to detect light intensity and automatically turn street lights ON or OFF.

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4. PWM Output

Definition:
PWM (Pulse Width Modulation) output pins generate a pulse signal that can simulate analog output. It is commonly used to control LED brightness and motor speed.

Example: LED Brightness Control

IoT Use Case:
A smart lighting system uses PWM to adjust the brightness of LEDs based on ambient light conditions.

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5. I2C/SPI Pins

Definition:
I2C and SPI are communication protocols used to connect Arduino with sensors, displays, memory modules, and other peripheral devices.

I2C Pins:

- SDA (A4)
- SCL (A5)

SPI Pins:

- MOSI (11)
- MISO (12)
- SCK (13)
- SS (10)

Example: OLED Display

IoT Use Case:
An IoT weather monitoring system uses I2C or SPI communication to connect temperature and humidity sensors with the Arduino.

Conclusion

Arduino UNO provides different types of pins such as Digital Input, Digital Output, Analog Input, PWM Output, and I2C/SPI communication pins. These pins help the Arduino interact with sensors, actuators, and communication devices, making it suitable for various IoT applications.